

Aaron Chakraborty

AI/ML Security Engineer | Systems Builder | B.Tech CSE

📍 Bhubaneswar, India 📞 8957622520 ✉ aaronchakraborty631@gmail.com in linkedin.com/in/aaron-chakraborty-309197287 🌐 github.com/aaron-631 🌐 aaron-631.github.io

Education

B.Tech Computer Science -K.I.I.T, Bhubaneswar CGPA: 9.42	2023-2027
Senior Secondary (ISC) -City Montessori School 94.25%	2023
High School (ICSE) – City Montessori School 98.25%	2021

Experience

AI/ML Research Intern	Mar 2026 – June 2026
SwiftSafe (Cybersecurity AI Startup)	Remote
- Owned an AI system end-to-end: designed and trained a 567M-parameter Mixture-of-Experts transformer from scratch (PyTorch, DeepSpeed ZeRO-2) – 16 experts (top-2 routing), GQA, RoPE, custom tokenizer. - Deployed a FastAPI inference service on A100 GPU infrastructure; built adversarial evaluation pipelines to continuously measure model robustness at production load. - Published technical write-ups on LinkedIn/X breaking down architecture and design trade-offs – translating deep ML decisions for both technical and non-technical audiences.	
Technology Apprentice (SEED)	Jun 2026 – Present
DBS Tech India	Remote
Selected for DBS's flagship SEED Apprenticeship from a highly competitive national pool; working on enterprise-scale digital transformation and infrastructure resilience inside a global financial institution.	
Lab Coordinator	Sep 2024 – Present
IoT Lab, K.I.I.T	K.I.I.T, India
Coordinated operations for 30+ active members; ran demos and technical showcases that helped drive student adoption of lab projects, acting as a liaison between the technical team and non-technical members.	
Security Testing Intern	May 2024 – Jul 2024
Panacea Infosec Pvt Ltd	Delhi, India
Ran 20+ vulnerability assessments and penetration tests and mapped findings to the CVSS score; authored 10+ technical risk reports translating attack-vector data into prioritized, business-level recommendations.	

Projects

Project Argus - AI Agent Risk Evaluation Platform	 GitHub
<i>Tech Stack: Python, asyncio, AES-256-GCM, Docker, YAML</i>	
- Architected a modular, async platform that statically analyzes AI agent configs, tool schemas, and permission scopes pre-deployment to catch risk before it ships. - Formulated a probabilistic risk-scoring model ($R = S_{\text{base}} \times C_{\text{env}} \times P_{\text{conf}}$) – a single data-driven metric distilled from multiple technical risk signals to drive go/no-go decisions.	
ReconForge - Reconnaissance Automation Toolkit	 GitHub
<i>Tech Stack: Bash, Python, Nmap, Rustscan, JSON</i>	
- Designed a modular automation toolkit that streamlines network and web enumeration, cutting manual toil by 60%. - Engineered dual-format (Markdown + HTML) reporting with structured JSON parsing, producing reproducible, time-stamped analytic artifacts.	
Search Arena - AI Algorithm Visualiser & Analytics Platform	 Live Demo  GitHub
<i>Tech Stack: React 18, Firebase, Vite, D3.js, CI/CD</i>	
- Built a full-stack visualisation of 6 classical AI search algorithms with animated playback and D3.js rendering to make optimization logic legible to non-experts. - Added a Firebase-backed real-time leaderboard tracking solve-time metrics; shipped via automated CI/CD (GitHub Actions to Vercel).	

Achievements

Service Selection Board (SSB) (AIR 83): TES-51 Merit List – national-level cognitive and leadership evaluation.  <i>Feb 2024</i> Pentathlon 2025 (AIR 45): Top 0.90% globally in a 48-hour national CTF – analytical problem-solving under pressure. <i>Apr 2025</i> eLearnSecurity Junior Penetration Tester (eJPTv2): INE Security, Score 91%.  <i>Feb 2026</i>	
--	--

Skills

AI, ML & Data: PyTorch, DeepSpeed, Transformer Architectures (MoE, GQA, RoPE), Training Pipelines, Predictive Modeling, Data Pipelines, FastAPI
Engineering, Communication & Risk: Python, SQL, Java, JavaScript, Bash, Docker, REST APIs, Git/CI-CD, Firebase, D3.js, Technical Writing, Risk Scoring, Adversarial/AI Evaluation, OWASP Top 10